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Home Automation System
Proposal Document

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1. BACKGROUND

In today's tech-driven world, smart home automation has become an important step toward improving safety, comfort, security, and energy efficiency. With affordable sensors and IoT-based microcontrollers, we can create smart systems that automatically monitor and respond to changes in the environment. This project focuses on designing a smart home automation system built around two microcontroller platforms working together: the **ESP32**, which provides Wi-Fi and Bluetooth connectivity for wireless monitoring and control, and the **Arduino Uno**, which independently manages outdoor lighting and a vehicle parking assistance feature.

The ESP32 side of the system uses three key sensors together with an RFID-based access control module to create a safer, more comfortable, and more secure home. The **MQ-5 gas sensor** detects flammable gases such as LPG, methane, and carbon monoxide and triggers a buzzer alert when concentrations become unsafe. The **DHT11 sensor** monitors kitchen temperature and humidity and automatically activates a ventilation fan when conditions exceed comfortable limits. An **RFID door lock system** replaces the conventional key with card-based authentication, unlocking the main door only when a registered RFID card or tag is presented. A **Blynk-based mobile application** displays live sensor readings and allows users to remotely switch home lighting on or off over Wi-Fi.

The Arduino Uno side of the system handles two further conveniences. An **LDR sensor** automates fence and rooftop lighting based on ambient brightness, while an **ultrasonic sensor** provides a simple parking assistance feature that alerts the driver before a vehicle gets too close to the garage wall. Overall, the system offers a cost-effective and user-friendly smart home solution that combines two complementary microcontroller platforms into a single, practical automation project.

2. PROBLEM AND MOTIVATION

2.1 PROBLEM

In modern homes, maintaining safety, comfort, security, and energy efficiency remains a significant challenge. Most households still rely on traditional, manually operated devices that lack automation and smart control. Many homes do not have gas detection systems capable of identifying dangerous gases such as LPG, methane, and carbon monoxide, which increases the risk of fire, explosion, or poisoning. Kitchens often experience rapid changes in temperature and humidity due to cooking, yet standard ventilation systems still depend on manual operation, leading to poor air quality and discomfort.

Home security is another growing concern. Traditional key-based door locks can be lost, duplicated, or picked, leaving homes vulnerable to unauthorised entry. Outdoor areas such as fences and rooftops typically rely on manually switched lighting, which wastes energy when left on during daylight hours or forgotten at night. In addition, reversing a vehicle into a garage without a clear reference point increases the risk of collision with the rear wall. Finally, the lack of remote monitoring and control makes it difficult for residents to manage their homes when away, particularly for adjusting lighting or checking on environmental conditions.

2.2 Motivation

The motivation for this project comes from the need for an affordable and integrated home automation system that addresses safety, comfort, security, and energy efficiency together. The system uses an **MQ-5 gas sensor** for early detection of flammable gases, a **DHT11 sensor** to maintain proper kitchen air quality through automatic ventilation, and an **RFID door lock** to provide secure, keyless entry. An **LDR sensor** automates outdoor lighting to reduce energy waste, while an **ultrasonic sensor** assists drivers during parking to prevent accidental collisions. A **mobile application** further allows remote monitoring of sensor data and control of home lighting. Overall, the project aims to provide an easy-to-use smart home solution that supports a safer, more comfortable, more secure, and more energy-efficient lifestyle.

3. AIM AND OBJECTIVES

3.1 Aim

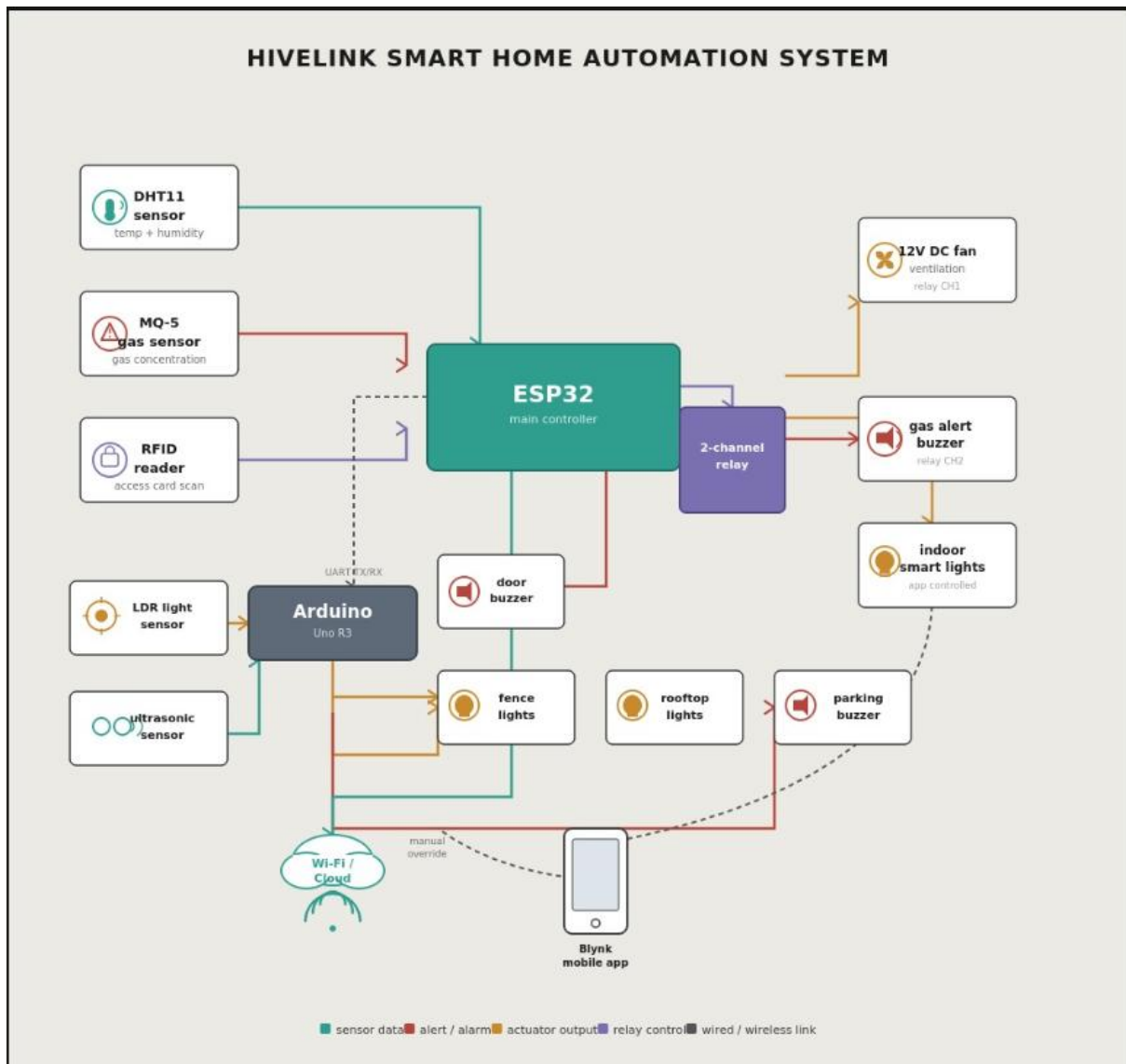
The main goal of this project is to create an affordable and reliable smart home automation system using **ESP32** and **Arduino Uno** technology. This system improves safety, comfort, security, and energy efficiency in daily life. It includes an MQ-5 gas sensor for early detection of gas leaks, a DHT11 sensor for monitoring kitchen temperature and humidity, an RFID-based door lock for secure keyless entry, an LDR sensor for automatic outdoor lighting control, and an ultrasonic sensor for parking assistance. The project also includes a mobile application that lets users remotely monitor sensor readings and control the lighting system. Overall, this project seeks to provide a user-friendly and cost-effective smart home solution by combining safety monitoring, environmental control, access security, and outdoor convenience features into one automated platform.

3.2 objectives

1. **System Design and Integration:** Design and connect all components, including the ESP32, Arduino Uno, MQ-5, DHT11, RFID module, LDR, ultrasonic sensor, relay modules, and buzzers. Test each component for proper operation and reliable communication.
2. **Gas Detection System:** Use the MQ-5 sensor to detect LPG, methane, and carbon monoxide gases, and activate a buzzer alarm when the reading exceeds the safe threshold.
3. **Kitchen Monitoring:** Use the DHT11 sensor to track temperature and humidity, and automatically trigger the ventilation fan when either value exceeds the defined safe limit.
4. **RFID Door Lock System:** Implement RFID-based authentication so that only registered cards or tags can unlock the main door, while unauthorised cards are rejected.
5. **Smart Outdoor Lighting Control:** Use the LDR sensor to automate fence and rooftop lighting based on ambient brightness, reducing unnecessary energy use.

6. **Parking Assistance System:** Use the ultrasonic sensor to measure the distance between a parked vehicle and the garage wall, and alert the driver through a buzzer as the vehicle approaches the safety threshold.
7. **Mobile Application Integration:** Develop a Blynk-based mobile application that displays temperature, humidity, and gas sensor values, shows gas status as safe or alert, and allows users to remotely control home lighting.
8. **System Testing:** Test the complete system under different environmental conditions to confirm sensor accuracy, fast response, and an overall reliability of about 95%.
9. **Refine and Debug:** Review test results, resolve any sensor, lock, or connectivity issues, and improve overall system performance for a better user experience.

4. SYSTEM DIAGRAM



5. METHODOLOGY

5.1 System Overview

Gas Detection System

a) Gas Detection

b) **a) Gas Detection** — The MQ-5 gas sensor continuously monitors the kitchen air for flammable gases such as LPG, methane, and carbon monoxide. It produces an analog signal proportional to the gas concentration and sends it to the ESP32 microcontroller.

c) **b) Activation Decision** — The ESP32 reads the analog signal and compares it against a preset threshold value of **2000**. If the reading exceeds this threshold, the system identifies a potential gas leak.

d) **c) Alarm Activation** — Once the gas concentration exceeds 2000, the ESP32 activates a buzzer to alert the occupants, allowing time for preventive action.

e) **d) System Reset** — When the gas concentration falls back below the threshold, the ESP32 automatically deactivates the buzzer and returns the system to standby mode.

Kitchen Temperature and Humidity Monitoring System

a) **Data Measurement** — The DHT11 sensor continuously measures kitchen temperature and humidity and transmits the readings to the ESP32 in digital format.

b) **Decision Making** — The ESP32 compares the readings against the defined thresholds of **30°C for temperature** and **60% for humidity** to determine whether conditions remain within safe and comfortable limits.

c) **Ventilation Control** — If the temperature exceeds 30°C **or** the humidity exceeds 60%, the ESP32 activates a relay that switches on a 12V DC ventilation fan to manage airflow and restore comfortable conditions.

d) **Automatic Deactivation** — When both temperature and humidity return to within the safe range, the ESP32 turns off the fan and resets the system to normal operating mode.

RFID Door Lock System

a) **Card Scanning** — The RFID reader, mounted at the main door, continuously scans for nearby RFID cards or tags presented by the user.

b) Authentication — When a card is detected, the ESP32 reads its unique identification number and compares it against a list of registered, authorised card identifiers.

c) Access Decision — If the scanned card matches a registered identifier, the ESP32 signals the lock mechanism to unlock the door for a short, predefined period before automatically re-locking. If the card is not recognised, access is denied and the door remains locked.

d) Security Benefit — By replacing the conventional key with card-based authentication, the system reduces the risk associated with lost or duplicated keys and improves overall home security.

Smart Lighting Control System

a) Light Intensity Detection — The LDR sensor, connected to the Arduino Uno, measures the surrounding light intensity and sends an analog signal to the microcontroller.

b) Decision Logic — The Arduino Uno compares the LDR reading against a preset brightness threshold of **100**. When the reading falls below 100, indicating low ambient light or night-time conditions, the system determines that lighting is required.

c) Lighting Control — When ambient light is low, the Arduino Uno switches on the fence and rooftop lights through a relay module. During daytime, when sufficient natural light is available, the lights are automatically switched off to conserve energy.

Parking Assistance System

a) Distance Measurement — The ultrasonic sensor, mounted on the garage wall, continuously measures the distance between itself and an approaching vehicle by emitting and receiving sound wave pulses.

b) Threshold Comparison — The Arduino Uno continuously compares the measured distance against a safety threshold of **2.9 cm**.

c) Alert Activation — When the vehicle reaches a distance of 2.9 cm from the wall, the Arduino Uno activates a buzzer to alert the driver and help prevent collision with the garage wall.

Mobile App Integration

The system connects to a **Blynk-based mobile application** via the ESP32's built-in Wi-Fi module. The app displays real-time temperature, humidity, and gas sensor values, and presents the gas status clearly as either **SAFE** or **ALERT** depending on the MQ-5 reading. In addition, residents can remotely switch the home lights on or off through the app from anywhere with an internet connection. This feature adds convenience, supports energy savings, and improves home security by allowing users to simulate occupancy while away.

5.2 Hardware Requirements

BREADBOARD — A solderless board for building and testing circuits. It connects the ESP32, Arduino Uno, sensors, and output devices with jumper wires, allowing for convenient testing and modification.

ESP32 MICROCONTROLLER — The ESP32 is the main controller for the kitchen safety, comfort, and access subsystems. It includes built-in Wi-Fi and Bluetooth, reads data from the MQ-5, DHT11, and RFID modules, and controls the buzzer, relay, and door lock mechanism.

ARDUINO UNO MICROCONTROLLER — The Arduino Uno independently manages the outdoor lighting and parking assistance subsystems. It reads input from the LDR and ultrasonic sensors and controls the corresponding relay and buzzer outputs.

MQ-5 GAS SENSOR — Detects gases such as LPG, methane, and carbon monoxide and sends analog signals to the ESP32, activating the buzzer when unsafe gas levels are present.

DHT11 SENSOR — Measures temperature and humidity in the kitchen. The ESP32 uses this data to control the ventilation fan when readings exceed the set thresholds.

RFID READER AND TAGS — The RFID module reads identification data from presented cards or tags. The ESP32 uses this data to authenticate users and control the door lock accordingly.

LDR SENSOR — Detects ambient light intensity and helps automate fence and rooftop lighting. The Arduino Uno turns the lights on in low-light conditions and off in bright conditions to save energy.

ULTRASONIC SENSOR — Measures the distance between a parking vehicle and the garage wall. The Arduino Uno uses this reading to trigger the parking assistance buzzer.

RELAY MODULES — Act as switches, allowing the ESP32 and Arduino Uno to safely control higher-power devices such as the ventilation fan, door lock, and outdoor lights using low-power control signals.

BUZZERS — Provide audible alerts for the gas detection system and the parking assistance system, ensuring a quick response from users.

LED LIGHTS — Energy-efficient outputs used for outdoor fence and rooftop lighting, automatically controlled by the Arduino Uno through the LDR sensor.

DOOR LOCK MECHANISM — An electronic lock, such as a solenoid or servo-driven latch, that is engaged or released by the ESP32 based on the outcome of RFID authentication.

JUMPER WIRES — Connect sensors, modules, and components on the breadboard for flexible, solderless circuit testing.

5.3 Software Requirement

Arduino IDE — The Arduino Integrated Development Environment (IDE) is free, open-source software for writing, compiling, and uploading code to both the ESP32 and Arduino Uno microcontrollers. It supports C and C++ programming and works across Windows, macOS, and Linux.

In this project, the Arduino IDE is used to program and control all components, including the MQ-5 sensor, DHT11 sensor, RFID module, LDR sensor, ultrasonic sensor, relay modules, and buzzers. It defines the logic for gas detection, kitchen ventilation, door access control, outdoor lighting, and parking assistance, and allows real-time monitoring of sensor data during testing.

Blynk IoT Platform — The Blynk platform provides the mobile application interface used to display sensor readings and control home lighting remotely. It connects to the ESP32 over Wi-Fi and provides a simple, customisable dashboard for monitoring and control.

5.4 Tools and Technologies

ESP32 — A powerful and flexible microcontroller with built-in Wi-Fi and Bluetooth, well suited to IoT-based home automation. In this project, the ESP32 acts as the main controller for the kitchen safety, comfort, and door access subsystems, processing data from the MQ-5, DHT11, and RFID modules to trigger alarms, control ventilation, and manage the door lock. Developed by Espressif Systems, it is known for high performance, low power consumption, and cost-effectiveness.

Arduino Uno — A widely used and beginner-friendly microcontroller board built around the ATmega328P chip. In this project, the Arduino Uno independently manages the outdoor lighting and parking assistance subsystems, processing data from the LDR and ultrasonic sensors to control lighting and trigger the parking buzzer. Its simplicity and strong community support make it well suited for straightforward sensor-and-output tasks.

RFID Technology — Radio-Frequency Identification (RFID) allows the system to identify a card or tag wirelessly when it is placed near the reader. This technology removes the need for physical keys and enables fast, reliable access control with minimal hardware.

Sensor Integration — The system uses five sensors across the two microcontrollers: MQ-5 for gas detection, DHT11 for temperature and humidity monitoring, RFID for access authentication, LDR for automatic outdoor light control, and an ultrasonic sensor for parking distance measurement. Together, these sensors enable real-time, automated decision-making throughout the home.

Wireless Communication — The ESP32's built-in Wi-Fi allows direct connectivity with the Blynk mobile application without requiring an external communication module, letting users monitor environmental conditions and control lighting from anywhere.

Open-Source Resources — Both the ESP32 and Arduino Uno platforms benefit from extensive open-source libraries and active developer communities, which simplify integration, accelerate development, and support troubleshooting throughout the project.

6. EVALUATION METHOD

The system evaluation looks at performance, accuracy, reliability, and user-friendliness to ensure safe, efficient, and comfortable operation in a smart home.

6.1 Functional Evaluation

Accuracy and Reliability: Test the MQ-5, DHT11, LDR, RFID, and ultrasonic sensors under various conditions to ensure accurate and stable readings. Verify that both the ESP32 and Arduino Uno process data consistently and respond correctly in real time.

Data Transmission and Connectivity : Confirm reliable communication between the ESP32 and the Blynk mobile application via Wi-Fi, ensuring accurate data transfer, no signal loss, and prompt execution of user commands

6.2 User-Centric Reliability

Installation and Maintenance: Assess how easily the system can be set up and used by regular users without technical expertise. Evaluate sensor and lock durability and the system's overall low maintenance requirements for long-term use.

Access Security: Verify that the RFID door lock correctly grants access only to registered cards and reliably denies unauthorised cards, ensuring consistent and secure operation.

6.3 Performance

Effectiveness: Measure the system's performance in detecting gas leaks, monitoring kitchen temperature and humidity, authenticating RFID access, controlling outdoor lighting, and assisting with parking. Test response times for alarms, the door lock, and lighting to ensure quick action.

Safety and Efficiency: Confirm that the MQ-5 provides instant alerts for gas leaks, the DHT11 ensures proper ventilation control, the RFID system grants access reliably, the LDR optimises outdoor lighting for energy savings, and the ultrasonic sensor gives a timely parking alert. Verify the mobile app's stability for reliable remote access, improving household convenience and security.

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